

Grégoire Pelletier

Mathematics and AI — Interpretability, Representation Learning, Deep Learning

E gregoire.pelletier80@gmail.com
W gregoirepelletier.github.io
gh [GregoirePelletier](#)
in [gregoire-pelletier1](#)

Research Interests

- Mechanistic interpretability of language models
- Neural architectures for sequential data: Transformers and State Space Models (Mamba), text and time series
- Local and embeddable models: small language models, deployment under limited resources, open-source ecosystem

Research & Professional Experience

- Mar 2026 – now **Machine Learning Research Intern, EDF R&D (SEQUOIA – E7S)**, Saclay
- Sparse Autoencoders for interpreting dense text embeddings on real industrial datasets
 - Latent space structure: superposition, alignment of learned directions with semantic concepts
 - Data-efficient interpretability pipelines under strong domain-data constraints
- Feb – Jul 2025 **Machine Learning Engineer, Paris Digital Lab**, La Défense
- LLM-based systems interfacing with structured APIs (SNCF)
 - Multimodal retrieval system (RAG) with visual document understanding; LLM-based knowledge extraction
- Jun – Nov 2024 **Machine Learning Intern, Emotors (Stellantis)**, Carrières-sous-Poissy
- Thermal estimator for electric machines using recurrent neural networks (RNN, PyTorch)
 - Physics-informed neural network modeling (Thermal Neural Networks) under embeddability constraints
- May – Jul 2023 **Statistical Research Intern, ENS Paris-Saclay (Centre Borelli)**, Gif-sur-Yvette
- Texture modeling with Gaussian fields and neural networks
- Feb – Jun 2023 **Supervised Research, Laboratoire de Mathématiques d'Orsay**, Orsay
- Random graphs and community detection via the Hashimoto matrix

Education

- 2022 – 2026 **Magistère of Mathematics & Master's in Mathematics and Artificial Intelligence**, Université Paris-Saclay, Orsay
- Optimization, deep and statistical learning, reinforcement learning, sequential/online learning, probability, measure theory
- 2019 – 2022 **Classe Préparatoire (Mathematics and Physics)**, Lycée Louis-le-Grand, Paris

Skills

- Programming Python, R, OCaml, SQL, LaTeX, Spark
- Machine Learning PyTorch, TensorFlow, Scikit-learn, Pandas, Hugging Face
- Languages French (native), English (C1 – Linguaskill), German